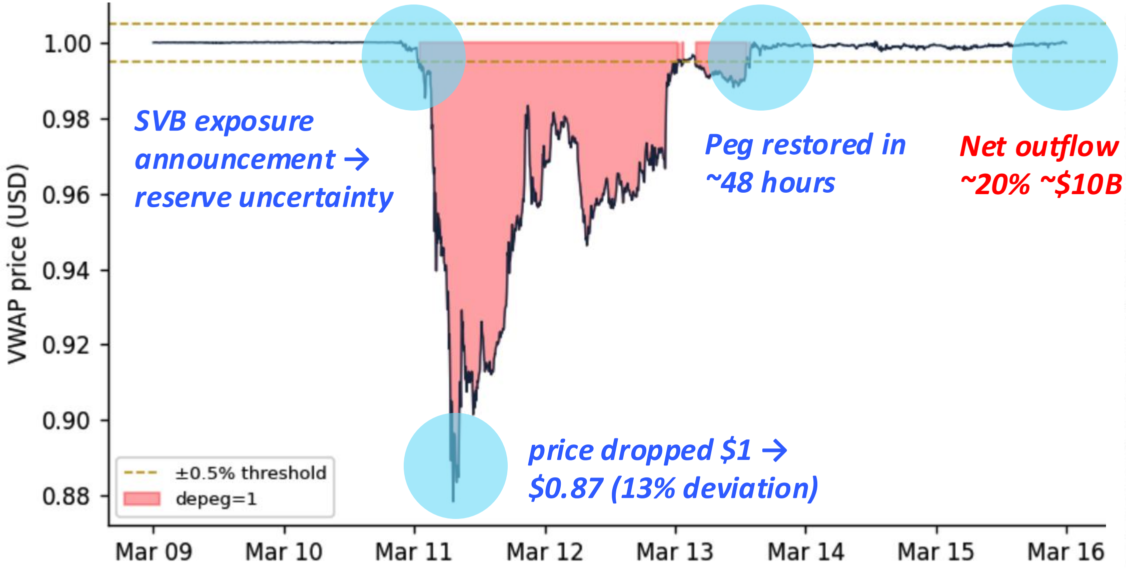
Predicting the Depeg

Machine Learning-Based Early Warning Systems Design

Depeg Is Temporary - The Cost Is Permanent

In theory: ***Volume Weighted Average Price = $1***



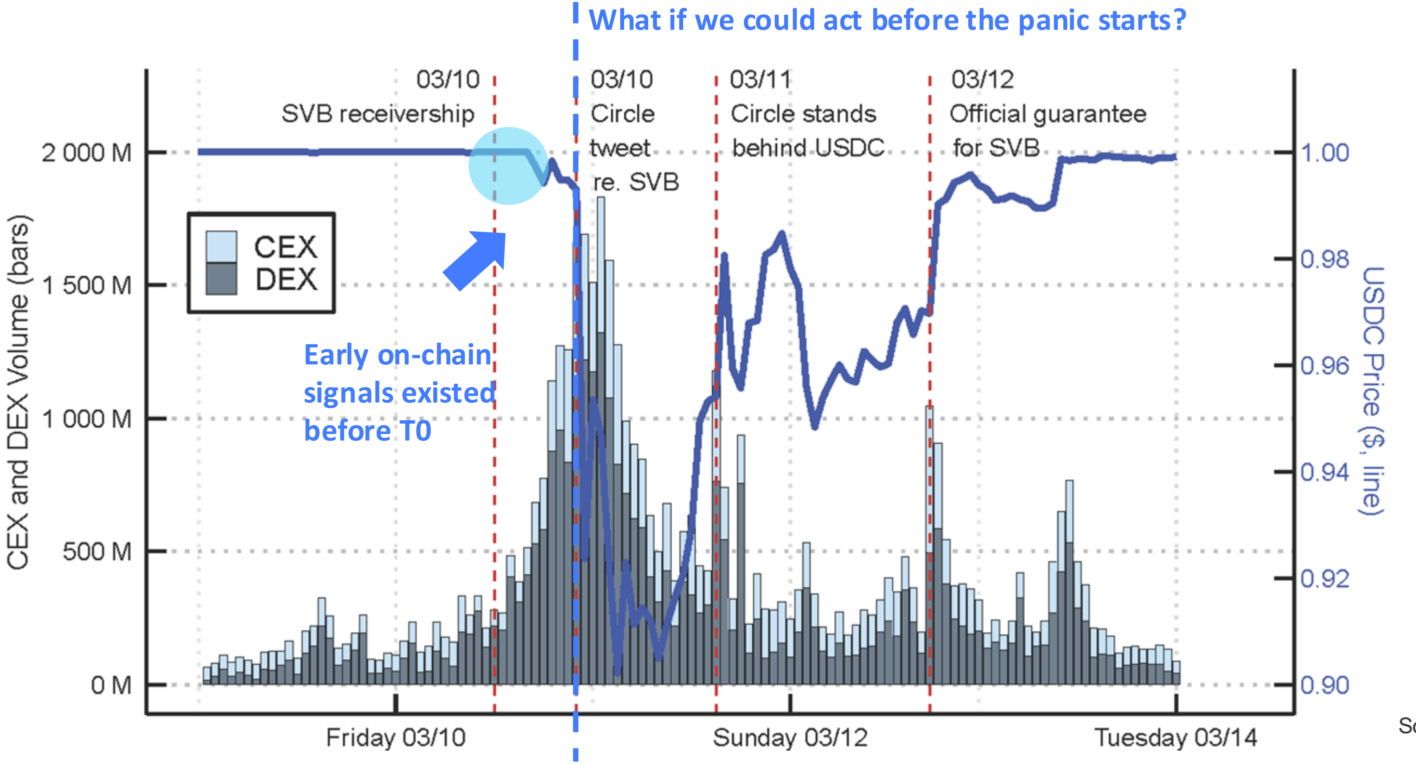
Stablecoins - Market Cap by Asset

(in $m)	3/8/23	3/20/23	Chg ($m)	Chg (%)
USDT	$71,722	$77,213	$5,491	7.7%
USDC	$43,227	$35,542	-$7,685	-17.8%
BUSD	$8,428	$8,228	-$200	-2.4%
DAI	$4,991	$5,418	$427	8.6%
TUSD	$1,269	$2,045	$776	61.2%
FRAX	$1,040	$1,050	$10	1.0%
USDP	$790	$812	$23	2.9%
USDD	$724	$728	$3	0.4%
GUSD	$555	$394	-$161	-29.0%
LUSD	$232	$265	$32	13.9%
USTC	$231	$223	-$8	-3.5%
MIM	$133	$108	-$24	-18.4%
SUSD	$123	$107	-$17	-13.6%
Other	$1,449	$1,139	-$310	-21.4%
Total	$134,915	$133,272	-$1,643	-1.2%

Data: DeFi Llama

*USDC recovered its peg in 48 hours — but Circle lost over **$1.35B** in economic value due to capital outflows*

From Reactive Crisis Handling to Proactive Risk Control



Source: CryptoCompare.

Our Project: Build a machine learning-based early warning system to detect instability signals and enable proactive intervention before a depeg escalates.

Project Overview

Jill Wang

Rob Springett

Ara Ludwig

Ify Enwezor

Secondary Research

- *Depeg Lifecycle*
- *Depeg Severity*
- *Cost of Depeg*

Problem Framing & Data Sourcing

- *Definition of Depeg*
- *On-chain data*
- *Market data*
- *Cross-chain flows*

Exploratory Data Analysis

- *Feature engineering*
- *Descriptive analytics*
- *Detect leading indicators*

Predictive Modeling

- *Early Warning Model*
- *A Three-Layer Defense System*
- *From Signal to Strategy*

01

Background Research

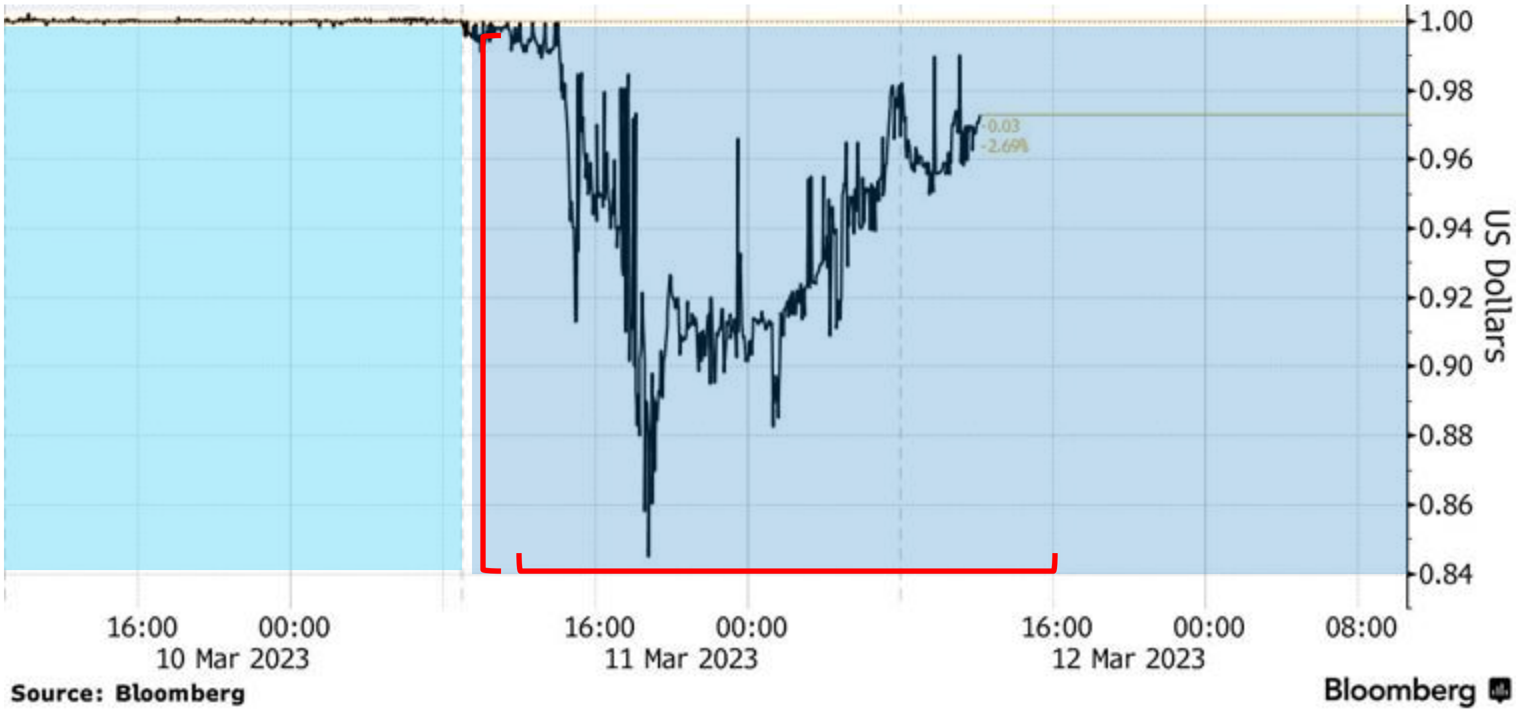
Early Conceptual Framework

Early Warning Stage

Depeg Stage

1.Will it depeg?

2.How bad will it be? How long will it last？



Defining & Measuring a Depeg Event

Early Warning Stage → **Depeg Stage**

Will it depeg?

Threshold choice drives everything downstream

Too tight → noise dominates labels
too loose → miss early-warning signals.

Duration filtering separates signal from noise

A 30-second flash crash on a thin exchange ≠ systemic stress.

Price source affects severity estimates

Single-exchange prices can overstate depegs by 5–10× vs. cross-exchange VWAP (Caramichael & Liao, 2022).

How bad will it be? How long will it last？

Severity Index = Depth × Duration

	<1h	1-6h	6-24h	>24h
<0.5%	*Level1*	*Level1*	*Level1*	*Level1*
0.5-2%	*Level1*	*Level1*	*Level2*	*Level2*
2-10%	*Level2*	*Level2*	*Level3*	*Level3*
>10%	*Level3*	*Level3*	*Level3*	*Level3*

Using Cost to Measure the Model's Business Value

Actual Loss for the Issuer

Depeg ➡ *Confidence ↓* ➡ *Supply ↓* ➡ *Reserves ↓* ➡ *Yield Income ↓*

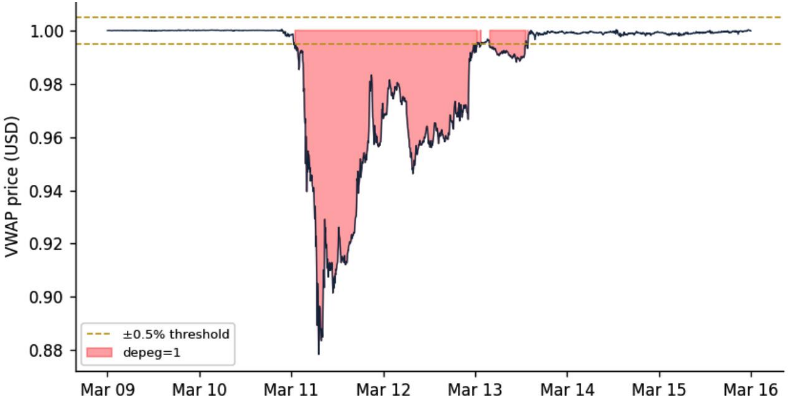
USDC Depeg Case	Value	Notes
Pre-Depeg Supply	~$43B	10-Mar-23
Post-Depeg Supply	~$32–33B	Within weeks
Net Outflow	**~$10–11B**	**Observed decline**
Average Yield on Reserves	~4.5% – 5%	US Treasuries (2023)
Time Horizon	1–3 years	Revenue impact period

Circle Loss: $10B * 4.5% * 3 = $1.35B

Net Outflow Estimation based on Severity

Severity	Outflow	Loss of Depeg in 3Y
Level1	5~10% * Supply	5~10% * Supply * Yield *3
Level2	10~20% * Supply	10~20% * Supply * Yield *3
Level3	20~60% * Supply	20~60% * Supply * Yield *3

USDC 2023 Depeg: Level 3 (13% Deviation, 48h)



Stablecoins – Chg in Market Cap by Asset & by Chain (3/8 – 3/20)

Source: Galaxy Digital Research

Stablecoins - Market Cap by Asset

(in $m)	3/8/23	3/20/23	Chg ($m)	Chg (%)
USDT	$71,722	$77,213	$5,491	7.7%
USDC	$43,227	$35,542	-$7,685	-17.8%
BUSD	$8,428	$8,228	-$200	-2.4%
DAI	$4,991	$5,418	$427	8.6%
TUSD	$1,269	$2,045	$776	61.2%
FRAX	$1,040	$1,050	$10	1.0%
USDP	$790	$812	$23	2.9%
USDD	$724	$728	$3	0.4%
GUSD	$555	$394	-$161	-29.0%
LUSD	$232	$265	$32	13.9%
USTC	$231	$223	-$8	-3.5%
MIM	$133	$108	-$24	-18.4%
SUSD	$123	$107	-$17	-13.6%
Other	$1,449	$1,139	-$310	-21.4%
Total	$134,915	$133,272	-$1,643	-1.2%

Stablecoins - Market Cap by Chain

(in $m)	3/8/23	3/20/23	Chg ($m)	Chg (%)
Ethereum	$79,796	$75,772	-$4,024	-5.0%
Tron	$38,147	$41,886	$3,738	9.8%
BSC	$7,009	$6,455	-$554	-7.9%
Polygon	$1,731	$1,599	-$132	-7.6%
Arbitrum	$1,469	$1,595	$126	8.6%
Solana	$1,819	$1,526	-$293	-16.1%
Avalanche	$1,492	$1,284	-$209	-14.0%
Optimism	$673	$695	$21	3.1%
Fantom	$438	$405	-$33	-7.4%
Omni	$252	$254	$2	0.8%
Algorand	$261	$237	-$24	-9.1%
Kava	$155	$152	-$3	-1.9%
Near	$99	$90	-$9	-8.8%
Other	$1,670	$1,444	-$226	-13.5%
Total	$134,912	$133,302	-$1,610	-1.2%

Data: DeFi Llama

02

Defining A Depeg Event

Our Process for Defining a Depeg Event

1

Literature Survey

Reviewed 12 academic papers, central bank working papers (BIS, Fed, IMF), and industry reports. Catalogued every depeg threshold definition and the rationale behind each choice.

2

Price Source Selection

Use a cross-exchange VWAP (IDX_REFRATE_VWAP). During SVB crisis, Coinbase showed USDC at $0.877 while the cross-exchange VWAP was $0.972 — single-exchange prices overstate severity (Caramichael & Liao 2022).

3

Threshold Calibration

Reviewed literature to see what worked and what didn't in previous studies. Adopted a flat threshold across all coins (consistent with BIS WP1111 and Fed SR1073).

4

Duration Calibration

Is there a time requirement for depeg events? We decided yes as arbitrage traders may not be able to execute in real time.

Key references: Bertsch BIS WP1111 · Anadu et al. Fed SR1073 · Lyons & Viswanath-Natraj (2023) · Nadler & Schär (2020) · Caramichael & Liao (2022) · Lee, Chiu & Hsieh (2025)

Our Stablecoin Depeg Definition & Modeling Label

Depeg Label Current Window = 1

if |price – \$1.00| ≥ 0.5% for ≥ 3 consecutive 5-minute bars (i.e. ≥ 15 min sustained)

- **Grounded in academic literature**

  ±0.5% is the standard threshold for fiat-backed stablecoins in central bank working papers (BIS WP1111; Federal Reserve SR1073).

- **Transitioning to a forward-looking target**

  The binary depeg flag tells us whether a coin is depegged right now — but our model needs to predict what will happen. We convert the label to a forward-looking binary. This shifts the question from detection to prediction, and matches the operational use case

Modeling Label - Depeg Next 4 Hours = 1

If any bar in the next 48 5-minute bars depeg label = 1

03

Data Sourcing Overview

Data Sources Overview

Eight sources across four categories — all unified at 5-minute resolution.

Peg Price Measurement

CoinAPI · IDX_REFRATE_VWAP

Cross-exchange volume-weighted average price at 5-minute resolution. The primary signal for whether a coin is depegged — aggregates across all major venues to eliminate single-exchange noise.

On-Chain Supply & Treasury Flows

Etherscan (ETH) · TronGrid (TRON) · Solana · XRPL · Omni

Mint, burn, and treasury transfer events across all chains. Captures institutional redemption pressure before it appears in open-market prices — over 2.5 million transfer events from 2017 to 2026.

On-Chain DEX Trading Activity

Curve Finance · 3pool · USDe/USDC · RLUSD/USDC · UST · BUSD pools

TokenExchange events on Curve — the dominant venue for large on-chain stablecoin swaps. Sell spikes reveal sophisticated traders exiting before open-market price breaks. 650,000+ swap events across five pools.

Market & Macro Context

Binance (BTC / ETH) · FRED (VIX, DXY, T10Y, FEDFUNDS) · Fear & Greed Index

Daily market and macro indicators forward-filled to 5-minute resolution. Provides the broader stress backdrop — a depeg during a VIX spike or crypto selloff carries very different signal from one in calm conditions.

Curve Pool Swaps — Why They Matter

What is Curve?

Curve Finance is the dominant on-chain exchange for large stablecoin swaps. Institutional traders and DeFi protocols use it to move millions between USDT, USDC, DAI and other stablecoins — often before those moves show up on centralised exchanges.

Why is it a leading indicator?

On-chain swaps are transparent and immediate. When sophisticated traders start selling a stablecoin on Curve, it reveals stress before the price breaks on open markets. These participants often have more information and act earlier than retail.

Analogy

Think of Curve like watching the "smart money" exit a position. A spike in USDT sell volume on Curve is the on-chain equivalent of institutional investors quietly moving for the exits before a price drops.

Key Signal

Curve Sell Spikes
→ Stable coin being swapped for other stablecoins
→ Sophisticated traders fleeing before open-market depeg

Pools Monitored

3pool	DAI / USDC / USDT	538k events
usde_usdc	USDe / USDC	106k events
rlusd_usdc	RLUSD / USDC	11k events
ust_3crv	UST / 3CRV	1.1k events
busd_3crv	BUSD / 3CRV	1.4k events

On-Chain Treasury Flows — Why They Matter

What are treasury flows?

When large institutions want to redeem stablecoins for real dollars, they send tokens directly to the issuer's treasury wallet on-chain (e.g. Tether's ETH or TRON treasury). We track every inflow and outflow to these known wallet addresses.

Why is it a leading indicator?

Institutions can redeem directly with Tether or Circle at exactly $1.00, bypassing the open market entirely. When redemption pressure spikes on-chain, it signals that large holders are exiting — often before the peg breaks on exchanges.

Analogy

Think of it like a bank run that starts quietly behind the scenes. Treasury inflow spikes are the on-chain equivalent of large depositors withdrawing funds before a bank failure becomes public news.

Key Signal

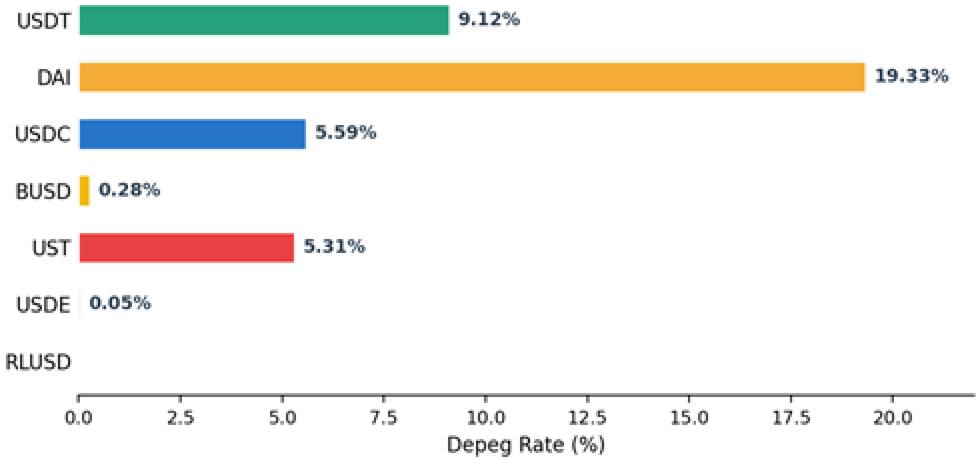
treasury_inflow spike
→ Institutions redeeming at par before market breaks
→ Leading indicator of open-market peg stress

Coverage		
USDT (ETH)	11.2k transfers	Nov 2017 → Feb 2026
USDT (TRON)	12.7k transfers	Apr 2019 → Feb 2026
USDC (ETH)	2.2M mint/burns	Sep 2018 → Feb 2026
DAI (ETH)	1.4M mint/burns	Nov 2019 → Feb 2026
RLUSD (XRPL)	5.6k mint/burns	Nov 2024 → Feb 2026
USDC (SOL)	711k mint/burns	Oct 2020 → Feb 2026
USDT (Omni/BTC)	862 transfers	Nov 2017 → Sep 2025

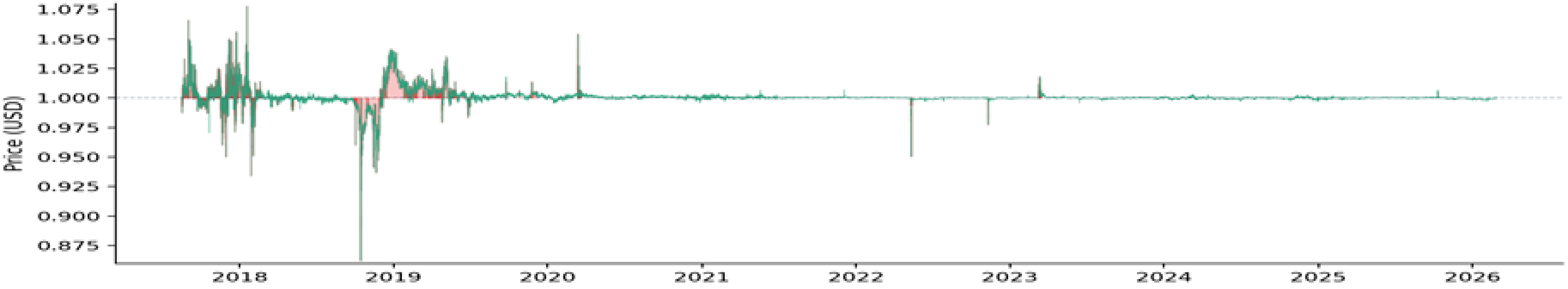
04

Exploratory Data Analysis

The Dataset: 7 Coins, 8.5 Years, Very Different Histories

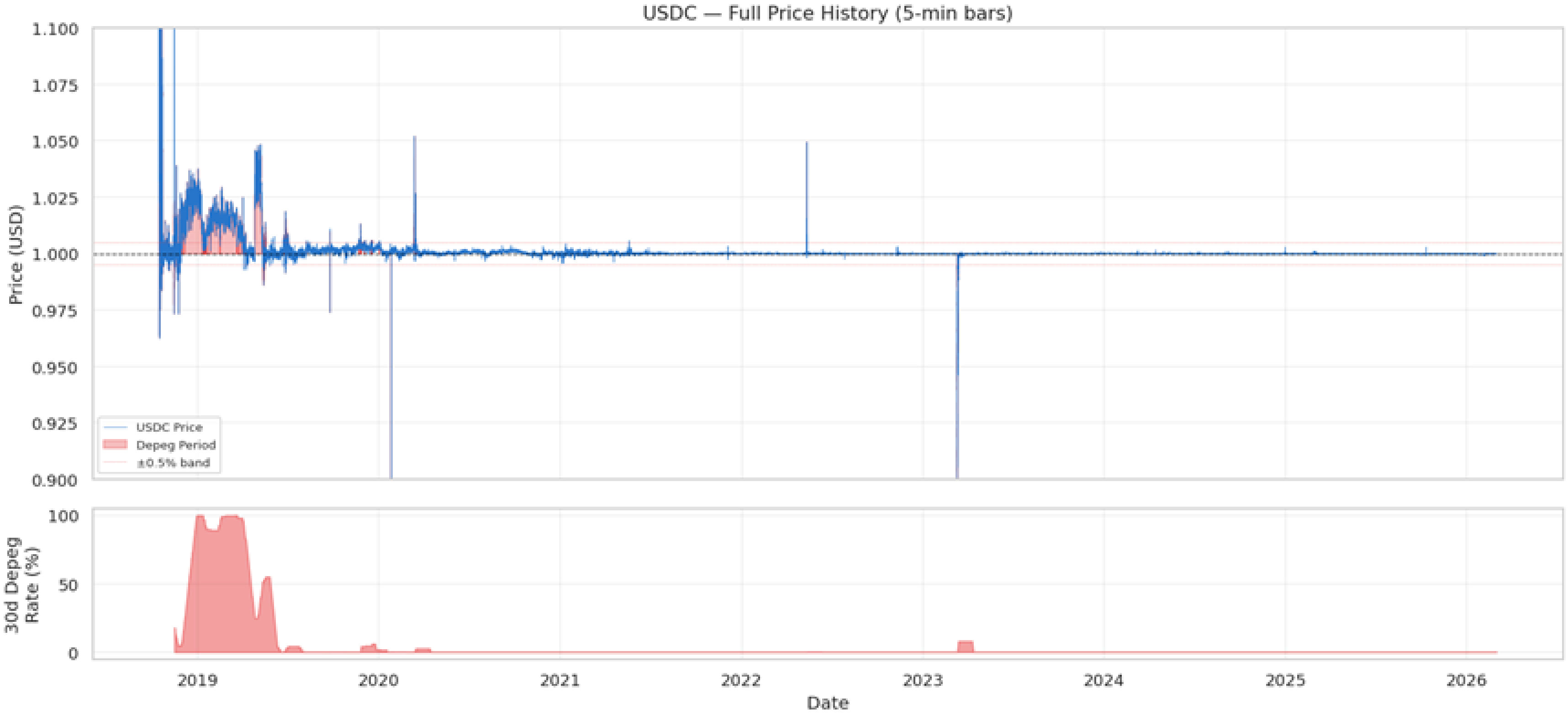


- Generated full statistical profiles (count, mean, std, percentiles, skewness, kurtosis) for all 54 numeric columns
- Checked for constant/near-constant columns. On-chain features (mint, burn) are 95%+ zeros but kept because rare non-zero values coincide with events
- Some missing data 21% of Binance data and 54% of depeg events, filled with neutral data



USDT price history (2017-2026). Red shading = depeg periods. Stress concentrates in 2017-2019; post-2020 is much more stable.

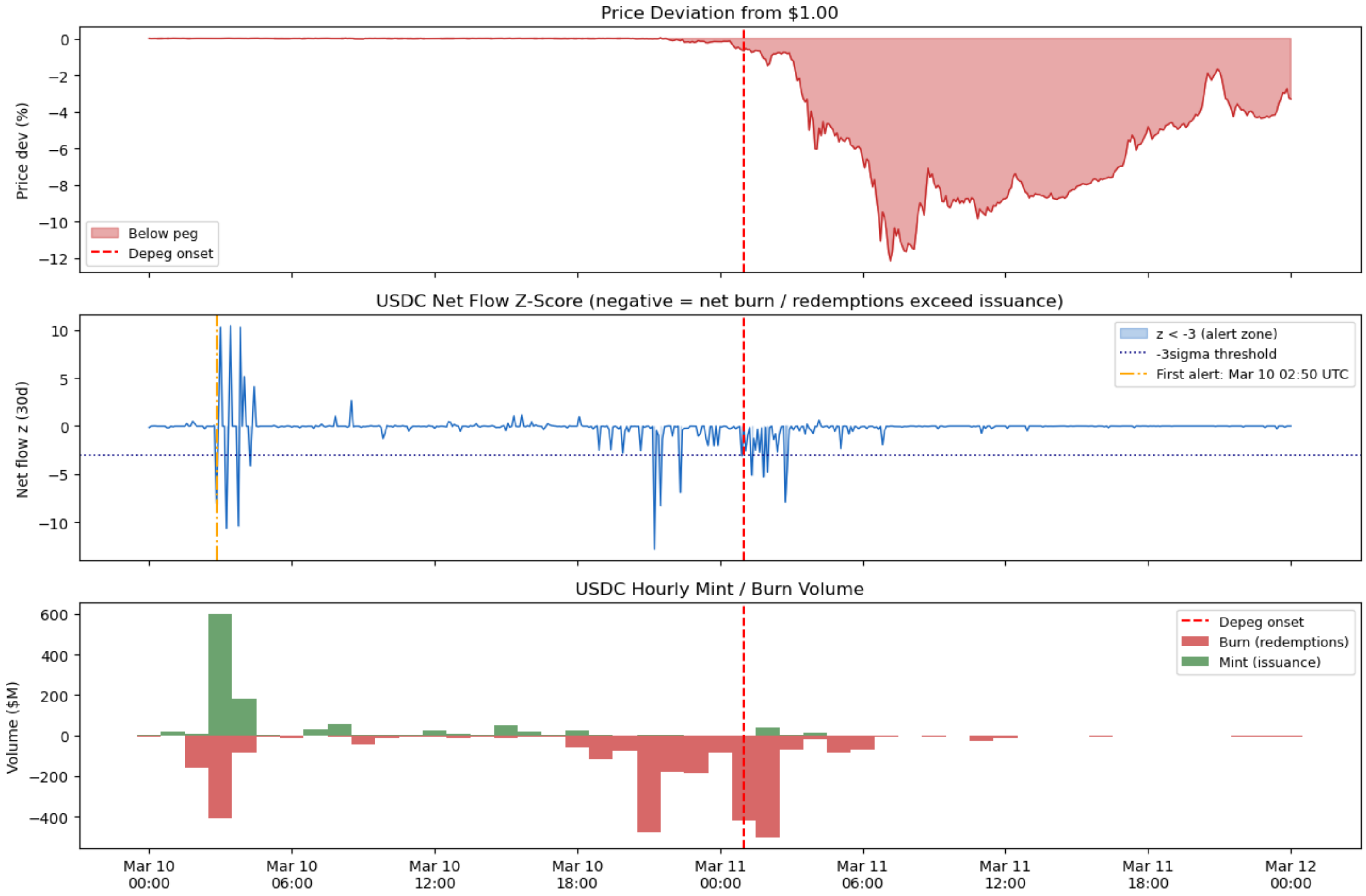
Case Study: USDC (Pricing History)



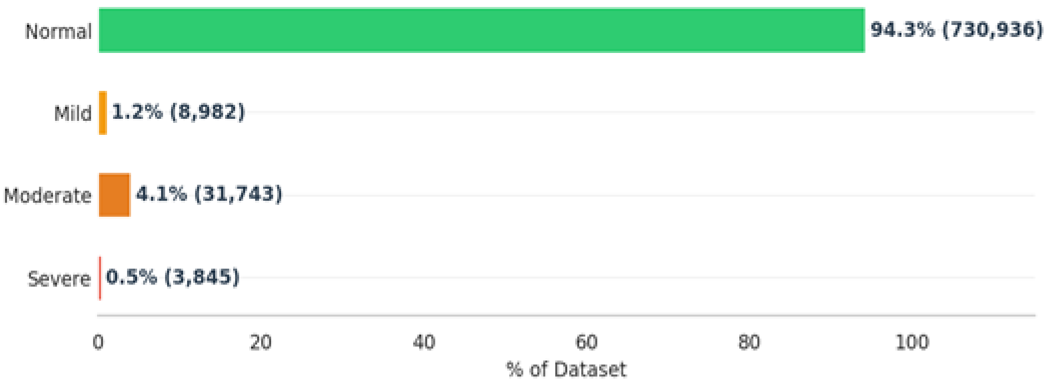
USDC price history (2018-2026). Red shading = depeg periods. Stress concentrates in 2018-2019; post-2020 is much more stable with some drops.

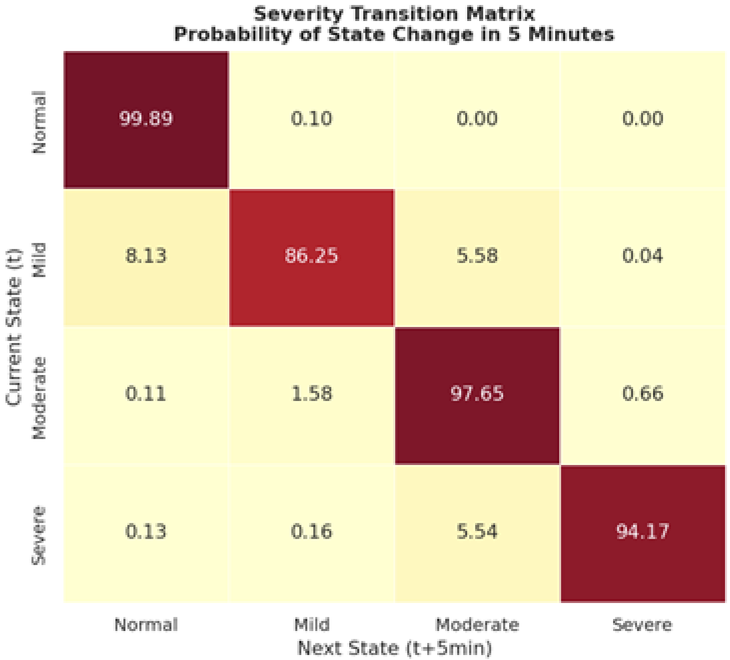
Leading Indicators before 2023 USDC SVB Depeg

Treasury and Mint/Burn volume volatility 24 hours before, dip around 3 hours before



Severity: USDC





4.1%
Moderate

>

1.2%
Mild

Escalation is gradual. Mild stays Mild 86% of the time. Only 5.5% chance of escalating to Moderate per 5-min bar. Direct jumps from Mild to Severe are near zero (0.04%). **This confirms the early warning system has a detectable window to alert before things get bad.**

05

Feature Engineering

Pooled Dataset — Class Balance & Composition

3,324,243
Total Rows

77
Columns

9.68%
Overall Depeg %

7
Coins Stacked

Coin	Rows	Depeg=1	Rate
USDT	897,923	89,107	**9.92%**
USDC	775,493	45,974	**5.93%**
DAI	828,950	175,521	**21.17%**
BUSD	370,835	1,414	**0.38%**
UST	153,948	9,530	**6.19%**
USDe	200,915	207	**0.10%**
RLUSD	96,179	0	**0.00%**

Modeling Notes

- RLUSD has no historical depeg events — learns entirely from pooled training
- DAI 21% rate reflects wider natural spread from crypto-collateral design
- Trailing 13 rows per coin dropped due to non number metrics
- Temporal train/val split required — no shuffling across coins
- Depeg (current state) included as future looking feature

9 Feature Groups — At a Glance

1 Price Deviation
11 features
1h · 4h · 1d rolling

2 Microstructure
5 features
spread · taker buy ratio

3 On-Chain Flows
~12 features
ETH · TRON · SOL · XRPL

4 Curve DEX
~8 features
6 pools · z-score

5 Market Returns & Vol
11 features
BTC/ETH ret + vol

6 Macro Levels
12 features
VIX · DXY · T10Y · FF · F&G

7 Temporal
5 features
hour · dow · session flags

8 Lag Features
8+ features
lags 1 · 3 · 6 · 12 bars

9 Cross-Coin
6 features
peer dev · peer depeg rate

06

Feature Selection

Feature Engineering Pipeline — Overview

7
Stablecoins

3.3M
Training Rows

68+
Feature Columns

9.68%
Overall Depeg Rate

Most Important Ind Features

Using Univariate AUC

- BTC Price
- Binance Volume
- Binance Quote Volume
- Take Buy Column / Quote
- Binance Trades
- ETH Price
- Spread Proxy
- US Dollar Index
- CoinAPI High

Feature Engineering

9 feature groups · 68+ features

- Rolling stats: 15m, 1h, 4h, 24h
- 30-day z-scores
- Lag features at 1/3/6/12 bars (5 min segments)
- Coin-specific flow aggregation
- Cross-coin systemic signals

Pooled Dataset

7 coins stacked

- 3,324,243 rows × 77 columns
- Target: Depeg in the next hour
- Class balance: 9.68% depeg
- Ready for ML training

Feature Selection — Reducing 68 → ~15 Features

1 Domain Pruning

Before any model

Manually remove near-duplicates using domain knowledge of feature groups

2 Correlation Filter

Threshold |r| > 0.95

Drop one of any pair with Pearson |r| > 0.95 (within-group collinearity)

3 SHAP Importance

SHAP values

Rank by mean |SHAP| across validation folds; cut at top 25–35 features

4 CV Validation

Confirm no AUC-PR loss

Re-train on reduced set; ensure AUC-PR within 0.5pp of full feature set

Expected Feature Survivors (~15)

Price
Price Dev · 1 hour SD Price Dev · Price Dev Difference · Instances over .03% (1 hour)

On-Chain
Normalized 30 day Treasury Flow·1 hour Burn total

Curve DEX
Normalized 30 day Curve Sell · 1 hour Curve Buy/Sell Ratio

Market
BTC Return (1 hour) · BTC 4 hour Volatility· 1 day VIX Difference

Lags
Price Dev (1 Lag)· Price Dev (6 Lag)

07

Predictive Modeling

In 2022, $40 billion vanished **in 72 hours.**

No risk model. No early warning. No exit window.

We built a system that reads the market's vital signs and rings the alarm before the collapse hits.

Think of it as a Bloomberg terminal alert but for stablecoin risk.

$40B+

UST/LUNA collapse · May 2022

23 hrs

Max lead time — our model (DAI)

3 of 5

Major crises detected early

How Do We Score the Early Warning?

Four metrics tell one story.

$$\text{Lift} = \frac{\text{Precision}}{\text{Base Rate}}$$

← % of alerts that were real depegs

← how often depegs naturally occur (~0.9%)

e.g. Lift = 67× means our alerts are 67× more likely to be real than a random guess

AUC-PR

How good across all risk thresholds?

*”On the days UST was about to collapse, our model had already moved it to the top of the danger list. A random guess scores 0.009, ours score 0.566 - **63x better**"*

Our result: 0.449 – 0.566

Lift (up to 67×)

How much better than random screening?

"Like Bloomberg watchlist, out of every 100 5-min bars our model flags as dangerous, 6 are real depegs (Precision). Without a model, you'd find less than 1 in every 1000."

Our result: 67× vs. random

Threshold

How sensitive is the trigger?

"Turn it down and every wobble triggers and alert, turn it up and you might sleep through the next UST. The sweet spot is where the model catches the most depegs

Our result: 0.12

F2 Score

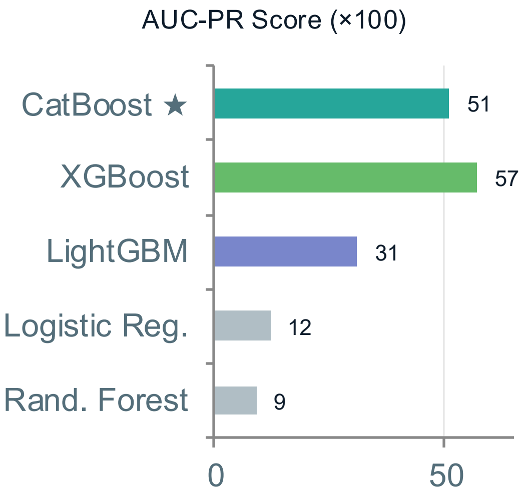
Did we catch the events that mattered?

"Missing the UST collapse cost investors $40B. A false alarm costs hedge. At 0.58, we caught most real events – deliberately erring on the side of caution"

Our result: 0.58 – 0.59

9 Models Evaluated. 3 Boosted Models Shortlisted.

Model	Native NaN	Class Imbalance	Ordered Boosting	Small Event Set†	Verdict
CatBoost ★	✓	✓	✓	~	Primary
XGBoost	✓	✓	✗	~	Alt.
LightGBM	✓	✓	✗	~	Alt.
Random Forest	✗	~	✗	~	Rejected
Logistic Reg.	✗	✓	✗	~	Rejected
LSTM / RNN	✗	✗	✗	✗	Rejected
LSTM Autoencoder	✗	✗	✗	✗	Rejected
Gaussian Mixture	✗	✗	✗	✗	Rejected
One-Class SVM	✗	✗	✗	✗	Rejected



★ Only CatBoost natively prevents leakage via ordered boosting with no extra tuning required.

+ Isolation Forest added as unsupervised layer to catch unknown-unknown events.

† Boosted models can handle small event sets with tuning (early stopping, class weights, learning rate).

MARCH 12, 2020 · BLACK THURSDAY · DAI DEPEG

How will our model have fared with the UST colapse?

ALEX — DeFi Retail Investor

- $50K in DAI · MakerDAO yield farming
- Holding since Jan 2020
- No early warning system

No model: ~$9,000 lost (18% of portfolio)

With model: exit before collapse. $9K saved.

Mar 12 · 00:00

🔕 Alex asleep. No alerts. Position fully exposed.

Mar 12 · 01:00

🤖 Model fires HIGH alert — AUC-PR 0.663 · Lead time: 23.2 hrs.

Mar 12 · 08:00

⚠️ DAI starts to depeg. Price slips to $0.91. Panic spreading.

Mar 12 · 18:00

🔥 DAI hits $0.82. Alex wakes up — position down $9,000.

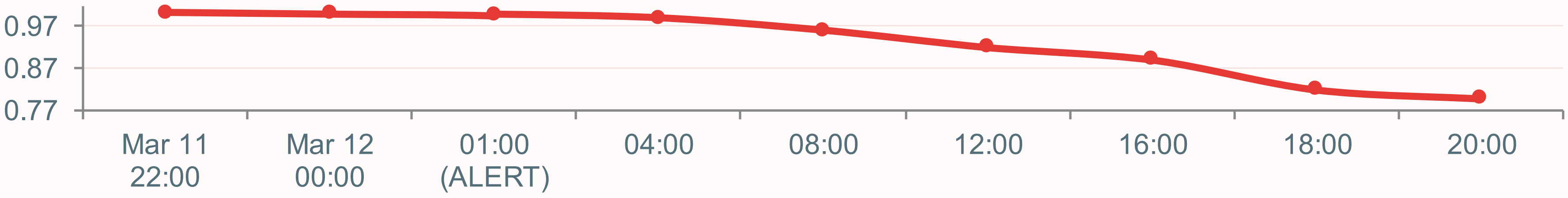
UST Risk Monitor

- Responsible for the $18B UST algorithmic peg
- Goal: protect the protocol

No model: $40B wiped – peg collapses to zero

With model: Limit Luna Dilution

DAI/USD Price — March 12, 2020 | Model alert fired at 01:00, giving Alex a 23-hour exit window



A Three-Layer Defense System

Each layer catches what the others miss

01 🔭 CatBoost — The Forecaster

Reads 20+ market signals and predicts instability up to 23 hours before it happens.

- Caught Black Thursday 23h early
- Validated on 5 real crisis events
- AUC-PR up to 0.566 · Lift up to 67×

LOEO validated:

Tested by pretending each crisis never happened then re-training and re-predicting. If it still catches the event, it truly generalises.

02 🚨 Isolation Forest — The Guard

Flags unusual market behavior that doesn't match normal market conditions.

- 0.28% precision-focused flag rate
- 1,272 anomaly flags in test data
- Trained on 'normal' only — no labels needed

LOEO validated:

Trained purely on normal market behavior. When the market goes abnormal in any way, this layer catches it even without historical precedent.

03 📋 SHAP — The Explainer

Tells you exactly which market signal triggered the alert so you act with confidence, not guesswork.

- Top signals: price momentum, volume surge
- Cross-asset stress (BTC/ETH correlation)
- Regulator-ready, auditable output

LOEO validated:

Every alert is paired with an explanation: 'Price momentum crossed -2.4σ, combined with a 340% volume spike.' No black box.

Recommendations for Improvements to the Model - Data Sourcing

GAP 1 — Order Book Depth

- Bid/ask spread and market depth at 5+ price levels
- Captures liquidity thinning before open-market depeg
- Spread widening often precedes price movement by 1–3 bars
- Sources: Kaiko, CoinAPI order book snapshots

GAP 2 — CEX Aggregate Flows

- Net stablecoin flows to/from centralized exchanges
- Large inflows signal intent to sell or redeem at scale
- Distinguishes on-chain DeFi activity from CEX redemption pressure
- Sources: Nansen, Glassnode, CryptoQuant (all paid)

Both gaps represent institutional behavior that likely leads open-market depeg by 5–30 minutes. Including either would strengthen predictive power at the 1-hour horizon.

From Signal to Strategy

Four recommendations for putting this system to work.

R1 Deploy as a Real-Time Risk Dashboard

Feed live 5-minute bar data into CatBoost + Isolation Forest. Output a 3-tier risk level (LOW / ELEVATED / HIGH) to portfolio managers and trading desks.

Immediate Implementation

R2 Integrate SHAP into Credit Risk Models

The top SHAP features (price momentum, volume z-score, cross-asset stress) can be embedded as dynamic risk factors in existing credit and liquidity models.

Risk Model Enhancement

R3 Use as Regulatory Stress Test Input

LOEO-validated performance on 5 historical crises provides a backtested, defensible signal that meets the explainability bar for regulatory review.

Compliance & Governance

R4 Expand Coverage — 7 to 20+ Coins

Current model covers 7 stablecoins. Extending to FRAX, LUSD, crvUSD and others increases systemic coverage and reduces blind spots in the broader DeFi market.

Future Roadmap

The question is no longer whether stablecoins can depeg. The question is will you know before it happens?

Thanks!